

Full solution packet for arXiv:2106.08949

A positive-area critical family is common hypercyclic

Status. full_solution_likely_valid. This packet gives a positive answer to Question 2.5 of Fernando Costa Jr., *Common hypercyclic algebras for families of products of backward shifts*, arXiv:2106.08949.

Source problem. For $a > 0$, put

$$w_s(a) = 1 + \frac{a}{\sqrt{s}}, \quad s \geq 1,$$

and let $B_{w(a)}$ be the associated weighted backward shift. Does there exist a positive-Lebesgue-measure set $\Lambda \subset \mathbb{R}^2$ such that

$$(B_{w(\lambda)} \times B_{w(\mu)})_{(\lambda, \mu) \in \Lambda}$$

has a common hypercyclic vector on $c_0 \times c_0$? The question appears on page 6 of the current arXiv PDF.

Example 2.4. Let $d \geq 1$, $X = \ell_p(\mathbb{N})$ or $c_0(\mathbb{N})$ and $(w(\lambda))_{\lambda > 0}$ be the family of weights defined by $w_n(\lambda) = 1 + \frac{\lambda}{n}$ (or $w_n(\lambda) := (1 + \frac{\lambda}{n})^\lambda$). Then the family $(B_{w(\lambda_1)} \times \cdots \times B_{w(\lambda_d)})_{\lambda \in (0, +\infty)^d}$ admits a common hypercyclic algebra for the coordinatewise product.

The problem of finding a common hypercyclic *vector* in the limit case $\alpha = 1/d$ remains open. Although we don't intend to explore this difficulty, it remains as an interesting open problem. We state it here in the form of a bi-dimensional example.

Question 2.5. Does there exist a set of parameters $\Lambda \subset \mathbb{R}^2$ with positive Lebesgue measure such that the family $(B_{w(\lambda)} \times B_{w(\mu)})_{(\lambda, \mu) \in \Lambda}$, with $w_n(\lambda) = 1 + \frac{\lambda}{\sqrt{n}}$, admits a common hypercyclic vector on $c_0(\mathbb{N}) \times c_0(\mathbb{N})$?

Proof intuition

The logarithm of the first n weights grows like $2a\sqrt{n}$. If a later parameter tag moves backwards by δ , the unwanted contribution has exponent of the form

$$C\delta\sqrt{n_j} - c\sqrt{n_j - n_k}.$$

Thus a backward move is harmless when $\delta^2 \lesssim (n_j - n_k)/n_j$. This is a *quadratic* cost.

A Hilbert space-filling curve has exactly the matching regularity: parameter time r^2 fills a square of side r . We partition its unit time interval into 4-adic pieces of length comparable to $1/n_j$, where the visit times n_j form an arithmetic progression. Their lengths have a divergent harmonic sum, so finitely many pieces cover the whole Hilbert interval. Hilbert's $1/2$ -Hölder estimate gives the quadratic cost above, while each image square has precisely the $n_j^{-1/2}$ size required by the common-hypercyclicity characterization.

The exact characterization

Set

$$f_n(a) = \sum_{s=1}^n \log \left(1 + \frac{a}{\sqrt{s}} \right), \quad f_0 = 0.$$

Theorem 2.1 of Bayart–Costa–Menet [2] applies to this family on any compact positive interval. Specialized to c_0^2 , it is enough to prove the following for every $\tau > 0$, $N \geq 1$, and $\varepsilon > 0$: there are finitely many tags $\lambda_h = (a_h, b_h)$ and integers

$$N \leq n_1 < n_1 + N \leq n_2 < \cdots < n_q$$

whose rectangles of side $\tau/\lceil\sqrt{n_h}\rceil$, ending at λ_h , cover the parameter set, and such that, for each coordinate,

$$\max_h e^{-f_{n_h}(a_h)} < \varepsilon, \quad (1)$$

$$\max_{j>k} \exp\left(f_{n_j+l}(a_k) - f_{n_j-n_k+l}(a_k) - f_{n_j+l}(a_j) + f_l(a_j)\right) < \varepsilon \quad (2)$$

for every $k < q$ and $0 \leq l \leq N$; the same inequalities hold with a_h replaced by b_h . These are exactly clauses (ii) and (iii) of that theorem because the displayed coefficients occupy distinct coordinates in c_0 .

The hypotheses of the characterization hold with $F(n) = \lceil\sqrt{n}\rceil$. Indeed, on $1 \leq a, b \leq 2$, differentiation and the mean-value theorem give

$$\frac{1}{3}\sqrt{n}|a-b| \leq |f_n(a) - f_n(b)| \leq 2\sqrt{n}|a-b|. \quad (3)$$

The pointwise weight ratios are also uniformly bounded away from zero.

A Hilbert scheduling lemma

We use a standard Hilbert map $H : [0, 1] \rightarrow [0, 1]^2$. There is an absolute constant $A \geq 1$ such that

$$\|H(s) - H(t)\|_\infty^2 \leq A|s - t|. \quad (4)$$

Moreover, every 4-adic interval of length 4^{-m} maps onto one dyadic square of side 2^{-m} .

Lemma 1 (Harmonic Hilbert schedule). *For every sufficiently small $\kappa > 0$, there is $L > 0$ with the following property. Given $\tau > 0$, an integer gap $G \geq 1$, and a sufficiently large initial integer, there are finitely many arithmetic times $n_h = n_0 + hG$, dyadic squares $Q_h \subset [1, 1 + L]^2$, and upper-right tags $\lambda_h \in Q_h$ such that*

- (i) *the Q_h cover $[1, 1 + L]^2$;*
- (ii) *the side of Q_h is at most $\tau/(2\sqrt{n_h})$;*
- (iii) *for every $k < j$,*

$$\|\lambda_j - \lambda_k\|_\infty \leq \kappa \sqrt{\frac{n_j - n_k}{n_j}}.$$

Proof. Choose $L > 0$ so small that $L^2 \leq \kappa^2/2$, and put

$$\beta = \min \left\{ \frac{\tau^2}{4L^2}, \frac{\kappa^2 G}{16AL^2} \right\} > 0.$$

Increase n_0 , if necessary, so that $\beta/n_1 < 1$. Starting at $t_0 = 0$, let $I_h = [t_{h-1}, t_h]$ be the next 4-adic interval whose length $\ell_h = 4^{-m_h}$ is the largest 4-adic number satisfying

$$\ell_h \leq \frac{\beta}{n_h}. \quad (5)$$

Because β/n_h decreases, the exponents m_h do not decrease. The current endpoint is therefore aligned with the next finer 4-adic grid. If it is below 1, the next interval fits. Maximality gives

$$\frac{\beta}{4n_h} < \ell_h \leq \frac{\beta}{n_h}. \quad (6)$$

If this procedure did not reach 1 after finitely many steps, the disjoint intervals would have total length at least $\frac{\beta}{4} \sum_h n_h^{-1} = +\infty$, a contradiction. Thus they form a finite partition of $[0, 1]$.

Let Q_h be the affine image in $[1, 1+L]^2$ of $H(I_h)$, and tag it by its upper-right corner λ_h . Its side is $L\sqrt{\ell_h} \leq \tau/(2\sqrt{n_h})$, proving (i)–(ii).

For $k < j$, the tag points have Hilbert parameters lying in I_k and I_j ; hence, by (4),

$$\|\lambda_j - \lambda_k\|_\infty^2 \leq AL^2 \sum_{h=k}^j \ell_h. \quad (7)$$

If $n_j/n_k \leq 2$, elementary comparison with an integral gives

$$\sum_{h=k}^j \frac{1}{n_h} \leq \frac{4}{G} \frac{n_j - n_k}{n_j}.$$

Equations (5) and (7), together with the definition of β , yield (iii). If $n_j/n_k \geq 2$, then $\|\lambda_j - \lambda_k\|_\infty^2 \leq L^2$, while $(n_j - n_k)/n_j \geq 1/2$; the choice of L again yields (iii). $\square$

Main result

Theorem 2. *There exists $L > 0$ such that, for $\Lambda = [1, 1+L]^2$,*

$$\bigcap_{(a,b) \in \Lambda} \text{HC}(B_{w(a)} \times B_{w(b)}) \neq \emptyset \quad \text{on } c_0 \times c_0.$$

In particular, Question 2.5 of arXiv:2106.08949 has a positive answer with a parameter set of positive Lebesgue measure L^2 .

Proof. On $[1, 2]$, the elementary bounds

$$f_m(a) \geq \frac{1}{3}\sqrt{m}, \quad |f_m(a) - f_m(b)| \leq 2\sqrt{m}|a - b| \quad (8)$$

hold for $m \geq 1$. The first follows from $\log(1+x) \geq x/(1+x)$, and the second from differentiation.

Fix

$$0 < \kappa < \frac{1}{12\sqrt{2}},$$

and let $L \leq 1$ be supplied by Lemma 1. We verify the exact characterization for the whole square Λ . Fix $\tau > 0$, $N \geq 1$, and $0 < \varepsilon < 1$. Put

$$C_N = \max_{0 \leq l \leq N, 1 \leq a \leq 2} f_l(a).$$

Choose an integer $G \geq N$ so large that

$$\exp\left(C_N - \frac{1}{6}\sqrt{G}\right) < \varepsilon. \quad (9)$$

Apply Lemma 1, enlarging its initial time so that $n_1 \geq N$ and

$$e^{-\sqrt{n_1}/3} < \varepsilon. \quad (10)$$

The rectangles ending at λ_h with side $\tau/\lceil\sqrt{n_h}\rceil$ contain Q_h , because $\lceil\sqrt{n}\rceil \leq 2\sqrt{n}$. They therefore cover Λ . The time gaps are $G \geq N$, and (10) together with (8) proves (1).

It remains to check the cross terms. Fix $k < j$, $0 \leq l \leq N$, and one coordinate of the tags. Writing that coordinate as a_h , the logarithm in (2) is at most

$$\begin{aligned} & 2\sqrt{n_j + N} |a_k - a_j| - \frac{1}{3}\sqrt{n_j - n_k + l} + C_N \\ & \leq 2\sqrt{2} \kappa \sqrt{n_j - n_k} - \frac{1}{3}\sqrt{n_j - n_k} + C_N \\ & \leq C_N - \frac{1}{6}\sqrt{n_j - n_k} \leq C_N - \frac{1}{6}\sqrt{G}. \end{aligned}$$

Here we used $n_j \geq N$ and clause (iii) of the scheduling lemma. The choice (9) proves (2); the other coordinate is identical. All clauses of Theorem 2.1 of [2] hold, so the family has a common hypercyclic vector. $\square$

Corollary 3. *The same square Λ works on $\ell_p \times \ell_p$ for every $1 \leq p < \infty$.*

Proof. For clause (ii) of the characterization, the p -th powers of the direct coefficients are bounded by $e^{-p\sqrt{n_h}/3}$, whose sum over the arithmetic progression becomes arbitrarily small when the initial time grows. For clause (iii), the proof above bounds the p -th powers of the cross coefficients by

$$e^{pC_N} e^{-p\sqrt{(j-k)G}/6}.$$

Their sum over $j > k$ becomes arbitrarily small as $G \rightarrow \infty$, uniformly in the finite truncation. These are precisely the relevant ℓ_p norms. $\square$

Verification, novelty, and limitations

The proof uses only the exact characterization in Bayart–Costa–Menet [2] and the standard recursive properties of the Hilbert curve. The two endpoint mechanisms are explicit: the 4-adic cells terminate because their lengths dominate a harmonic series, and the 1/2-Hölder bound turns Hilbert time into the quadratic parameter jump needed by the weight ratios. No numerical assumption is used.

On 17 August 2026, the run registry, ledger, attempt index, and local arXiv source corpus were searched for arXiv:2106.08949 and the exact positive-area question. Exact-phrase and close weighted-shift web searches found the source paper and unrelated common/frequent hypercyclicity work, but no later answer. The closest related source located was Costa’s 2024 paper on self-similar fractals [3], which concerns critical fractal parameter sets of zero area. This is a bounded novelty check, not a priority guarantee.

Human-review recommendation. Review as a claimed full positive answer, strengthened from c_0 to all classical ℓ_p spaces. The main points to audit are the variable-scale 4-adic alignment in Lemma 1 and the exact cross-ratio calculation in the proof of Theorem 2.

References

- [1] F. Costa Jr., *Common hypercyclic algebras for families of products of backward shifts*, arXiv:2106.08949, Question 2.5.
- [2] F. Bayart, F. Costa Jr., and Q. Menet, *Common hypercyclic vectors and dimension of the parameter set*, Indiana Univ. Math. J. **71** (2022), 1763–1795; arXiv:2103.13152, Theorem 2.1.
- [3] F. Costa Jr., *Self-similar fractals and common hypercyclicity*, J. Funct. Anal. **287** (2024), 110473; arXiv:2306.16026.